

Erick Alejandro Carrillo Lopez

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B.S. Computer engineering

University of Guadalajara

Began in **January - 2021**

Expected Graduation **December - 2026**

- **GPA:** 91/100
- **Notable Courses:** Data Structures, Algorithms, Database, Statistics and Numerical Analysis, Compiler and Translation, Theory of Computation, Computer networks, Computer Architecture, Distributed Computers

Work Experience

Software Development Engineer Intern - Amazon

- **May 2025 - May 2026** · Guadalajara, Mexico Metropolitan Area · On-site
- Engineered a **90% reduction in UI latency** for 2,000+ monthly active users, cutting worst-case response times from 10s to 1s by deprecating a monolithic **GraphQL** query and implementing lazy loading.
- Maintained core platform pages using **React (TSX)** and **Cloudscape Design System**, architecting reusable components and leveraging **useMemo**, **useRef**, and **useCallback** to resolve rendering bottlenecks.
- Optimized backend data retrieval with **API pagination** for AWS services, reducing payload sizes and resolving **Sev-2** incidents caused by **AWS Lambda** 6MB limits.
- Developed and maintained serverless infrastructure using **Java**, **AWS Lambda**, and **DynamoDB** for a scalable, cost-effective architecture.

SW Test Engineer Intern - Toshiba Global Commerce Solutions

- **Jul 2024 - Apr 2025** · Guadalajara, Mexico Metropolitan Area · On-site
- Develop and maintain automated test scripts using **Cypress**, **Selenium** with **Java** for web applications.
- Implement **Behavior-Driven Development (BDD)** methodologies with **Cucumber** to define and execute test scenarios, integrated with **Allure** for reporting.
- Collaborate with cross-functional teams to gather and understand requirements for effective test case design.
- Perform continuous integration and continuous deployment (**CI/CD**) tasks to ensure the automated test suite is integrated into the development **Jenkins** pipeline.

Projects

Memory Allocator in C:

- Developed **mem-c**, a custom memory allocator in C using a binary heap data structure.
- Achieved efficient memory block search with **$O(\log n)$** worst-case time complexity.
- Integrated Linux **mmap** syscall for dynamic memory allocation.
- Implemented support for memory pagination and adjacent chunk merging.
- Designed with future enhancements in mind, including garbage collection and arenas.

Technical Skills

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- **Back-End:** Python, C/C++, Java, Rust, Clisp, Flask and Django.
 - **Front-End:** JavaScript, HTML, CSS, Bootstrap, ReactJs and Tkinter.
 - **DataBase:** MySQL, MariaDB, PostgreSQL, SqlServer and Firebase.
 - **Developer Tools:** Git, Emacs, Linux environment, CMake and Makefile.
 - **Languages:** Spanish(Native), English(Basic).

Extracurriculars

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- **Clubs:** "Club de Algoritmia CUCEI" (Algorithms club).
 - **Awards or Honors:** Acknowledged for exceptional contributions as part of the "**Eureka**" team, representing the **University of Guadalajara CUCEI**, at the **18th National Programming Contest** in October 2023.